

Vendor Performance Report

Executive Summary

This report looks at vendor performance across 10,692 records. It highlights the main metrics, market trends, and practical steps to improve vendor relationships and profitability.

1. Problem Statement

Business Context

The company is in beverage distribution and works with many vendors. It needs a clear view of purchase and sales performance, pricing, and logistics costs.

- **Vendor selection and relationships** — find the best vendors and those that need attention
- **Profitability** — understand margins and pricing
- **Inventory management** — track stock turnover and sales speed
- **Cost control** — spot freight and logistics issues

Key Questions Addressed

1. Which vendors contribute most to overall sales and purchases?
2. What are the profit margins and gross profit distributions?
3. How efficient is inventory turnover across products?
4. Which brands drive the highest sales volume?
5. Are there pricing anomalies or inefficiencies?
6. What is the relationship between order size and unit purchase price?
7. Which products should receive promotional attention?

2. Data Overview

Dataset Composition

- **Total Records:** 10,692 vendor-product combinations
- **Time Period:** Complete inventory cycle (begin to end inventory)
- **Key Metrics:** 18 performance indicators tracked per vendor-product pair

Key Attributes Analyzed

- **Pricing:** Purchase Price, Actual Price (Selling Price)
- **Volume:** Product Volume (bottle size), Purchase Quantity, Sales Quantity
- **Financial:** Purchase Dollars, Sales Dollars, Gross Profit, Profit Margin
- **Operational:** Stock Turnover, Sales-to-Purchase Ratio, Freight Cost, Excise Tax
- **Categorical:** Vendor Name, Brand, Product Description

3. Exploratory data insights

Summary Statistics

Metric	Count	Mean	Std Dev	Min	25%	50%	75%	Max
VendorNumber	10,692	10,651	18,754	2	3,951	7,153	9,552	201,359
Brand	10,692	18,039	12,662	58	5,793.5	18,761.5	25,514.25	90,631
PurchasePrice (\$)	10,692	24.39	109.27	0.36	6.84	10.46	19.48	5,681.81
ActualPrice (\$)	10,692	35.64	148.25	0.49	10.99	15.99	28.99	7,499.99
Volume (ml)	10,692	847	664	50	750	750	750	20,000
TotalPurchaseQuantity	10,692	3,141	11,095	1	36	262	1,976	337,660
TotalSalesQuantity	10,692	3,077	10,953	0	33	261	1,929	334,939
TotalPurchaseDollars (\$)	10,692	30,107	123,068	0.71	453	3,655	20,738	3,811,252
TotalSalesDollars (\$)	10,692	42,239	167,655	0	729	5,298	28,397	5,101,920
TotalSalesPrice (\$)	10,692	18,794	44,953	0	289.71	2,857.80	16,059.56	672,819
TotalExciseTax (\$)	10,692	1,774	10,976	0	4.80	46.57	418.65	368,242
GrossProfit (\$)	10,692	12,132	46,224	-52,003	53	1,400	8,660	1,290,668
ProfitMargin (%)	10,692	-inf	NaN	-inf	13.32	30.41	39.96	99.72
FreightCost (\$)	10,692	61,434	60,938	0.09	14,070	50,294	79,529	257,032
StockTurnover (ratio)	10,692	1.71	6.02	0	0.81	0.98	1.04	274.5
SalestoPurchaseRatio	10,692	2.50	8.46	0	1.15	1.44	1.67	352.9

Note: -∞ appears due to zero/negative revenue scenarios; filtered in cleaned data

4. Data Quality

Key Data Quality Issues Identified

Negative and Zero Values

- **Gross Profit:** The lowest value is -\$52,002.78, which means some transactions lost money.
- **Profit Margin:** The lowest value is $-\infty$, which happens when revenue is zero or lower than cost.
- **Total Sales Quantity and Dollars:** Zero values mean some items were bought but never sold.

Outliers and Wide Ranges

- **Purchase and Actual Prices:** The highest prices are much larger than the average, showing a small number of premium items.
- **Freight Cost:** Costs range widely from 0.09 to 257,032, which suggests inconsistent logistics spending.
- **Stock Turnover:** Values range from 0 to 274.5, showing that some items move very quickly while others stay in stock.

Cleaned Dataset

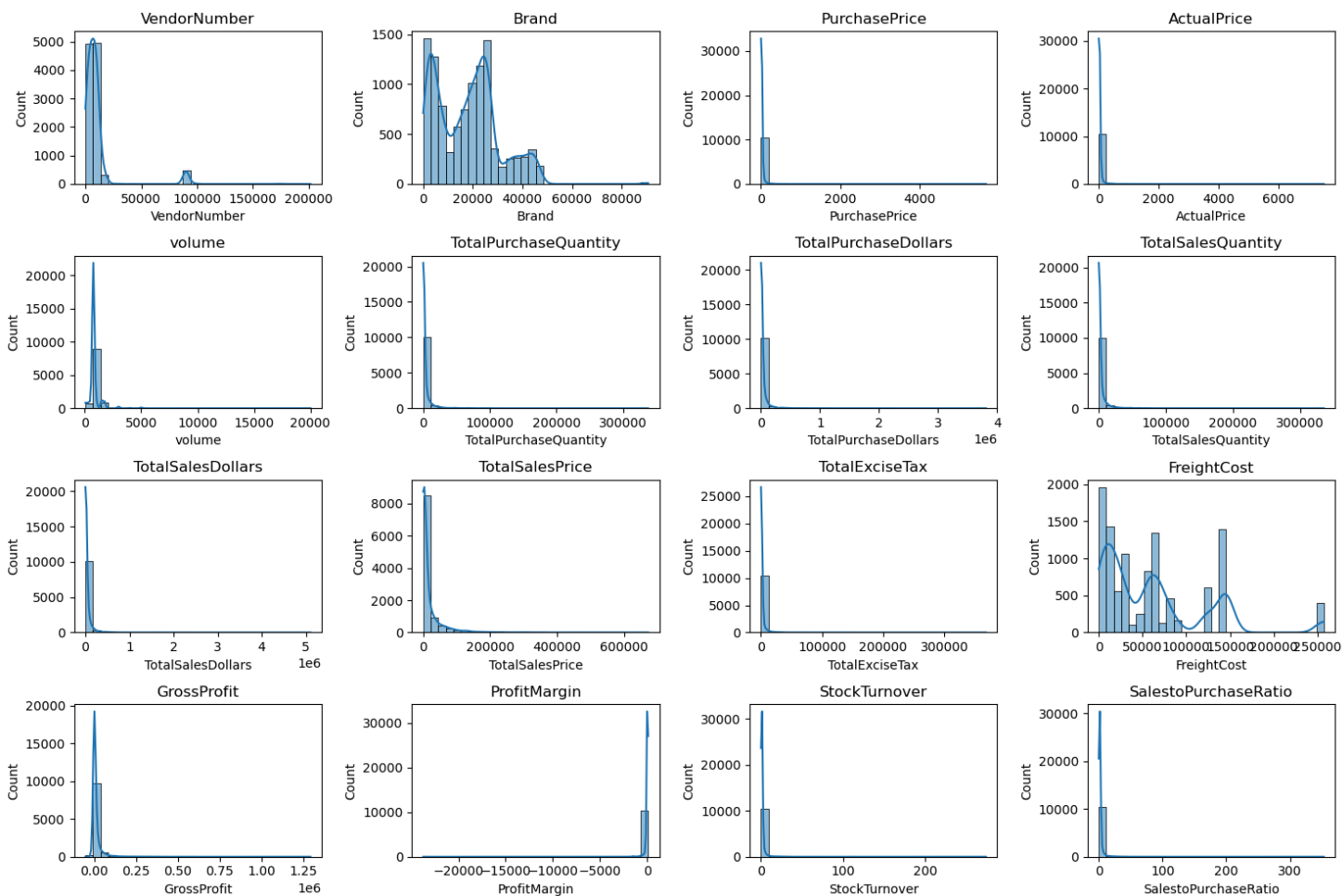
After filtering ($\text{GrossProfit} > 0$, $\text{ProfitMargin} > 0$, $\text{TotalSalesQuantity} > 0$), the dataset focuses on profitable, sold items for a clearer business review.

5. Data Distributions & Visualizations

5.1 Distributions - All numerical columns

The histogram below shows the distribution for the numeric columns. It highlights:

- **Right-skewed data:** most numbers are low, with a few very high values
- **Profit margin behavior:** this metric looks closer to a bell curve
- **Multiple groups:** some variables show more than one cluster of values

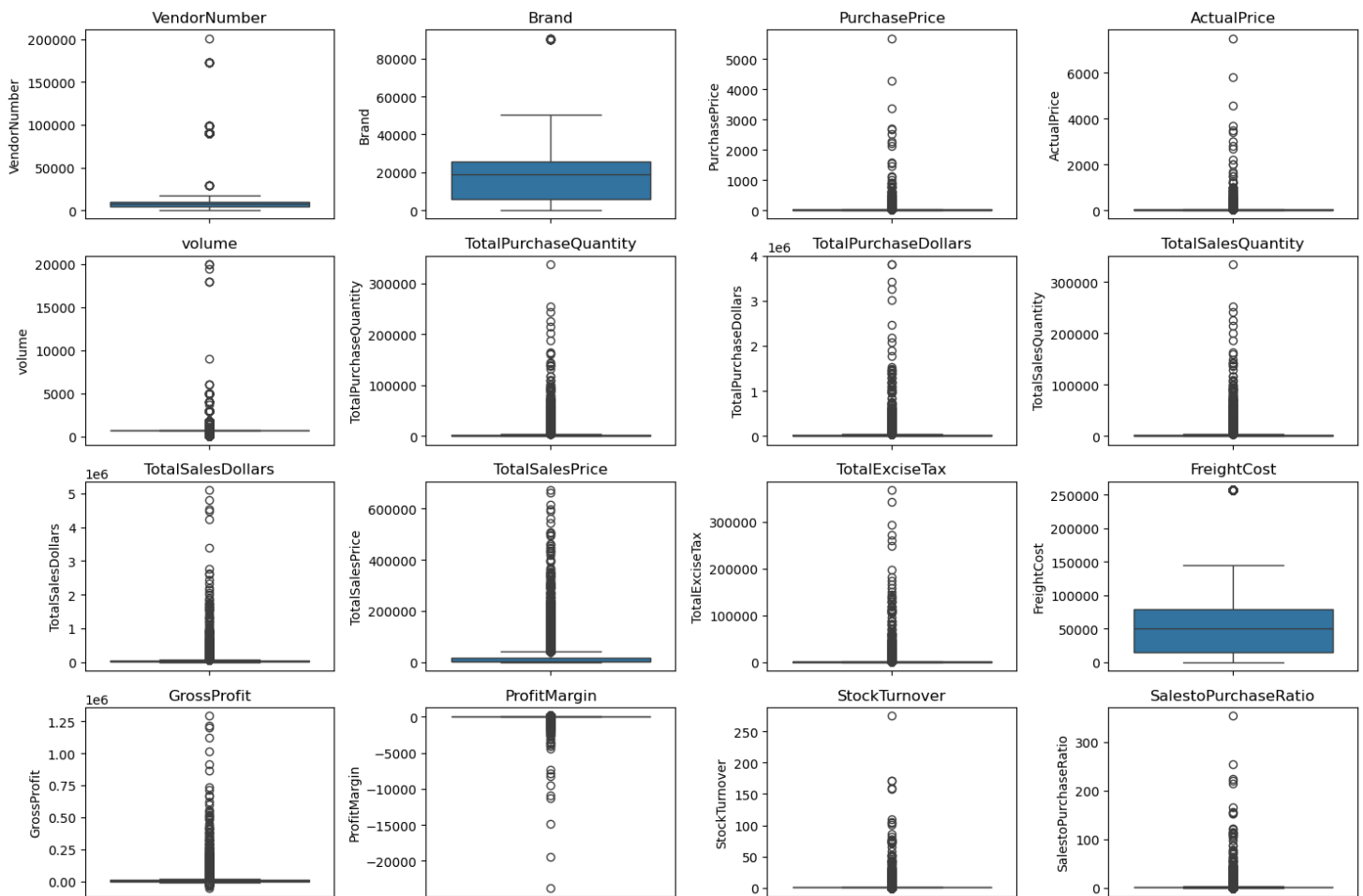


Key Observations:

- Most purchase prices cluster below \$1,000, with few high-value items
- Sales quantities and dollars follow similar right-skewed patterns
- Stock turnover concentrates around 0.98 (near 1:1 purchased-to-sold ratio)

5.2 Outlier Detection - Box Plots

The box plot shows extreme values and how spread works:

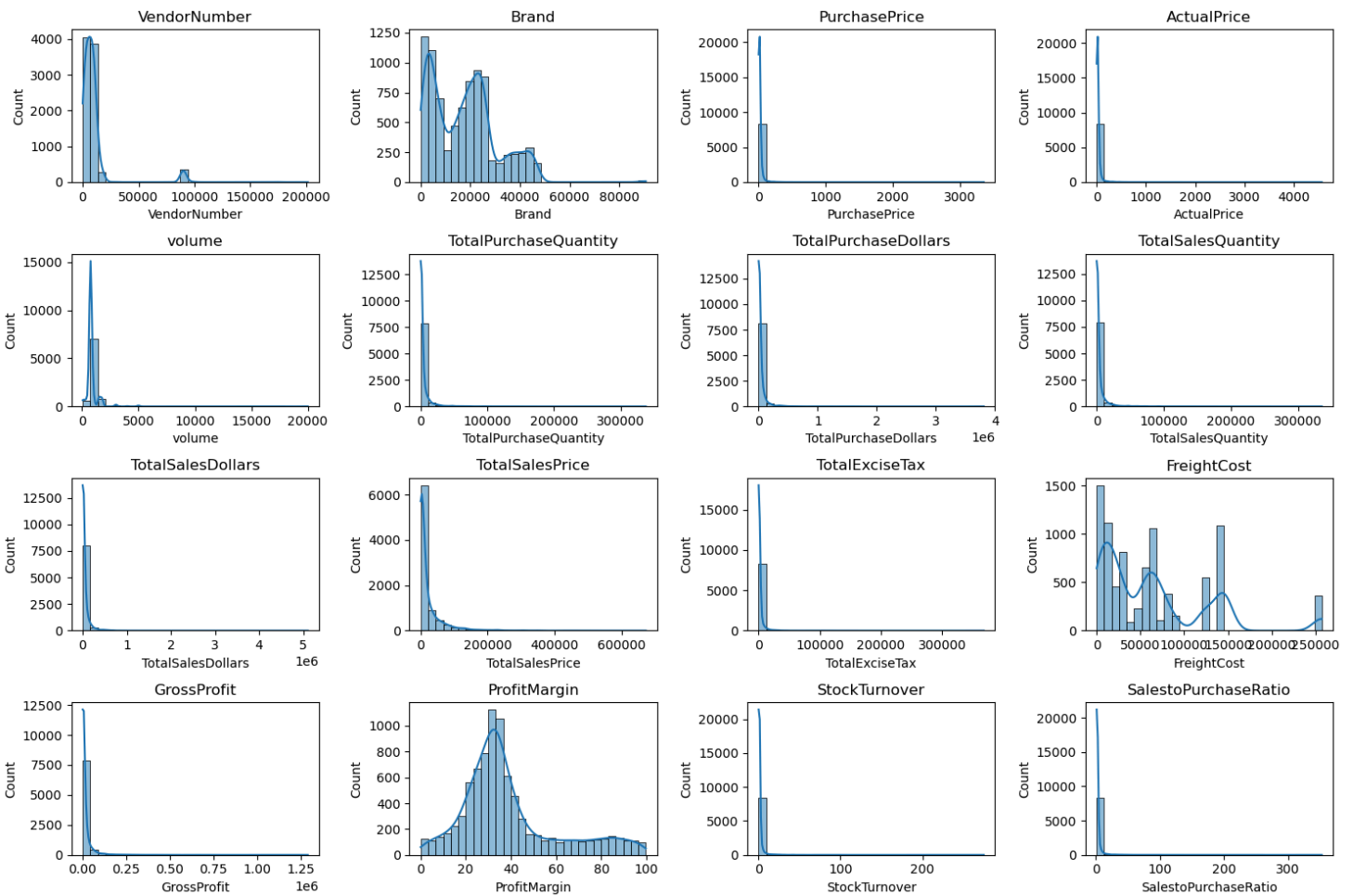


What stands out:

- **Freight Costs:** There are extreme outliers, which suggests the logistics cost can be very inconsistent.
- **Purchase and Actual Prices:** There are several very high values, showing a premium product group.
- **Stock Turnover:** Some items have unusually high turnover values above 100.
- **Sales-to-Purchase Ratio:** Ratios over 100 may signal data issues or old stock being sold.

5.3 Cleaned Data Distribution (Profitable Products Only)

After removing loss-making transactions, distributions normalize:

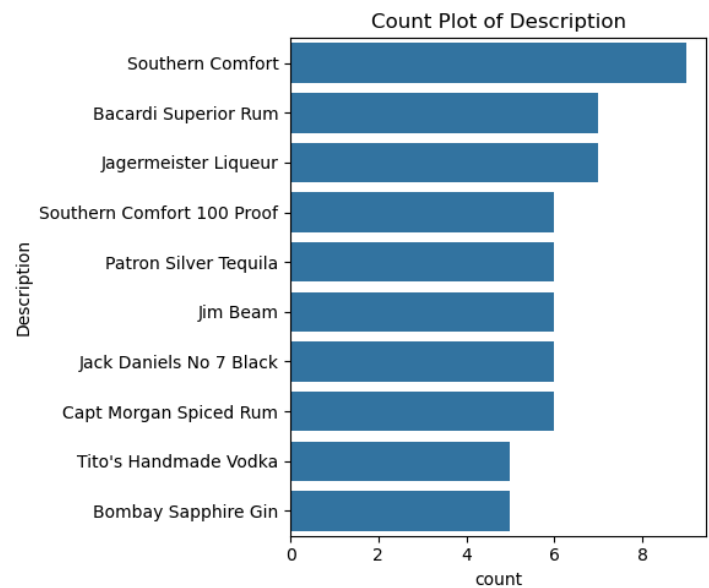
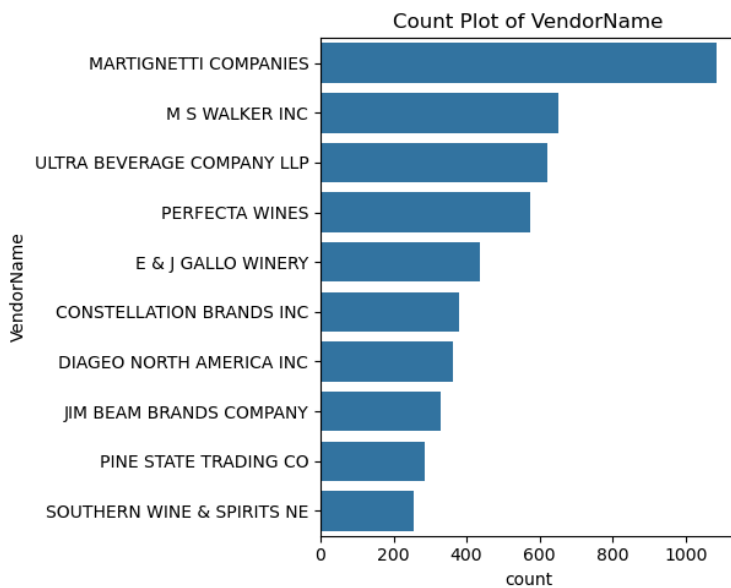


After filtering:

- Profit margin looks more centered around 30-40%
- The numbers behave more like normal patterns
- Outliers are still there, but the data is easier to interpret

5.4 Vendor & Product Mix

Vendor and product counts show market concentration:



Top Vendors by Volume:

- MARTIGNETTI COMPANIES: ~1,000 product-vendor combinations
- M S WALKER INC: ~600 combinations
- ULTRA BEVERAGE COMPANY LLP: ~550 combinations

Top Brands by Volume:

- Southern Comfort: 8+ product variants
- Bacardi Superior Rum: 6+ variants
- Jägermeister Liqueur: 6+ variants

5.5 Correlation

The heatmap shows how the variables relate to each other:

Strong links (>0.80):

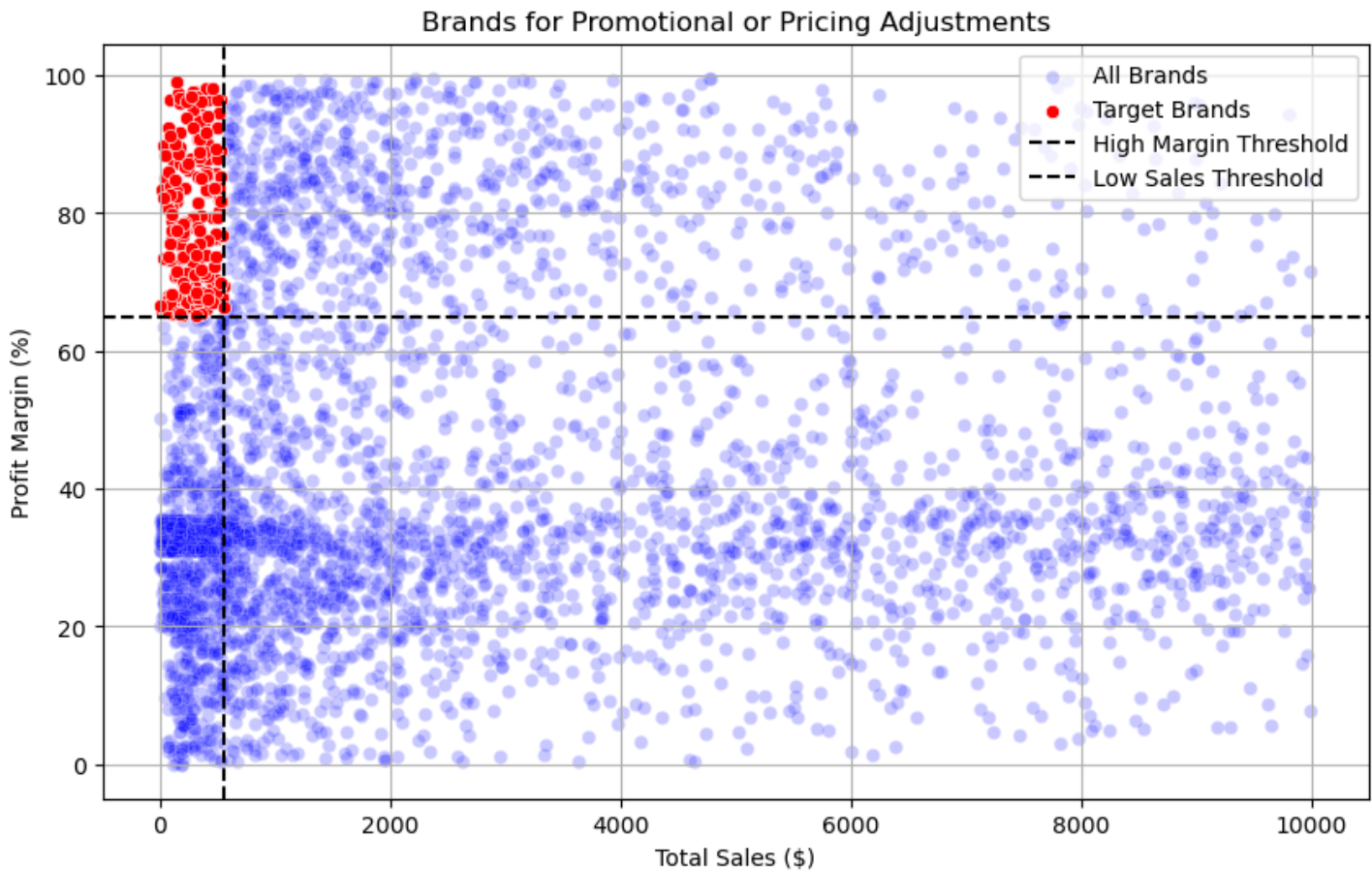
- TotalPurchaseQuantity ↔ TotalSalesQuantity (0.99): Purchases and sales move together closely
- TotalPurchaseDollars ↔ TotalSalesDollars (0.82): Dollar volume follows quantity patterns
- PurchasePrice ↔ ActualPrice (0.99): The markup is consistent
- TotalExciseTax ↔ GrossProfit (0.87): Bigger sales usually mean bigger tax and profit

Weaker links:

- FreightCost shows low correlation with most metrics (0.14-0.20)
 - This suggests freight is not tied closely to order size and may be more fixed or inconsistent.
- ProfitMargin ↔ SalestoPurchaseRatio (0.40-0.42): A small relationship, not very strong

6. Performance review

6.1 Promotional and pricing opportunities

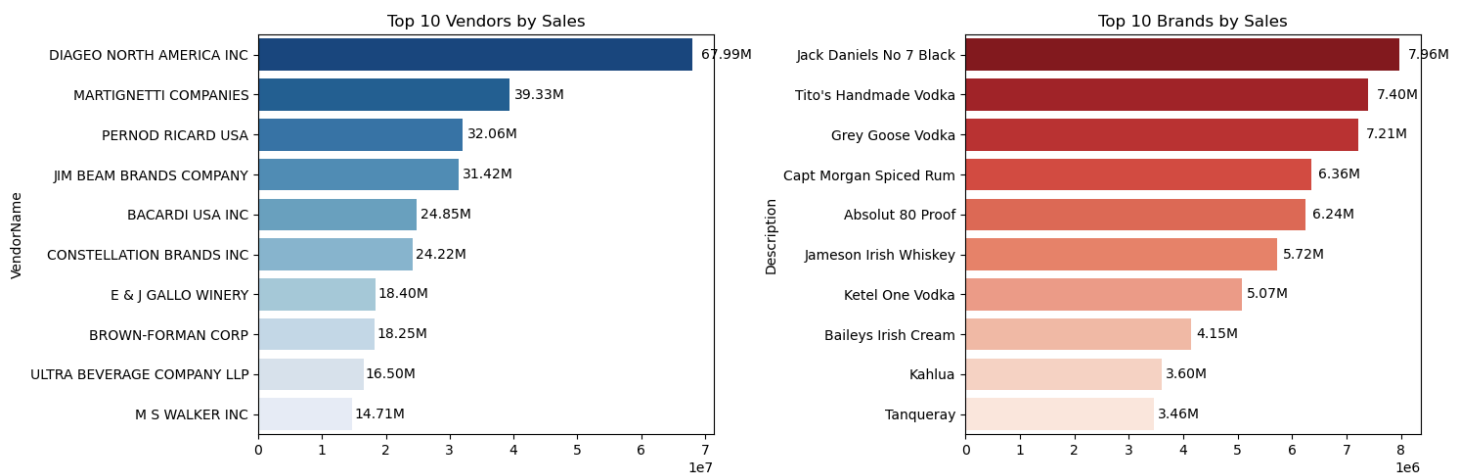


Profit margin vs. total sales:

- **Red dots** (target brands): High margin ($>65\%$) and low sales ($<\$500$)
 - Suggested action: consider promotions or small price cuts to improve volume
 - About 75 brands fit this pattern
- **High margin line:** 65% profit margin
- **Low sales line:** \$500 total sales

A good next step is to promote high-margin products that have low sales, without cutting too much into profit.

6.2 Sales Leaders - Top 10 Vendors & Brands



Top Vendors by Total Sales

Rank	Vendor	Sales
1	DIAGEO NORTH AMERICA INC	\$61.99M
2	MARTIGNETTI COMPANIES	\$39.33M
3	PERNOD RICARD USA	\$32.06M
4	JIM BEAM BRANDS COMPANY	\$31.42M
5	BACARDI USA INC	\$24.85M
6	CONSTELLATION BRANDS INC	\$24.22M
7	E & J GALLO WINERY	\$18.40M
8	BROWN-FORMAN CORP	\$18.25M
9	ULTRA BEVERAGE COMPANY LLP	\$16.50M
10	M S WALKER INC	\$14.71M

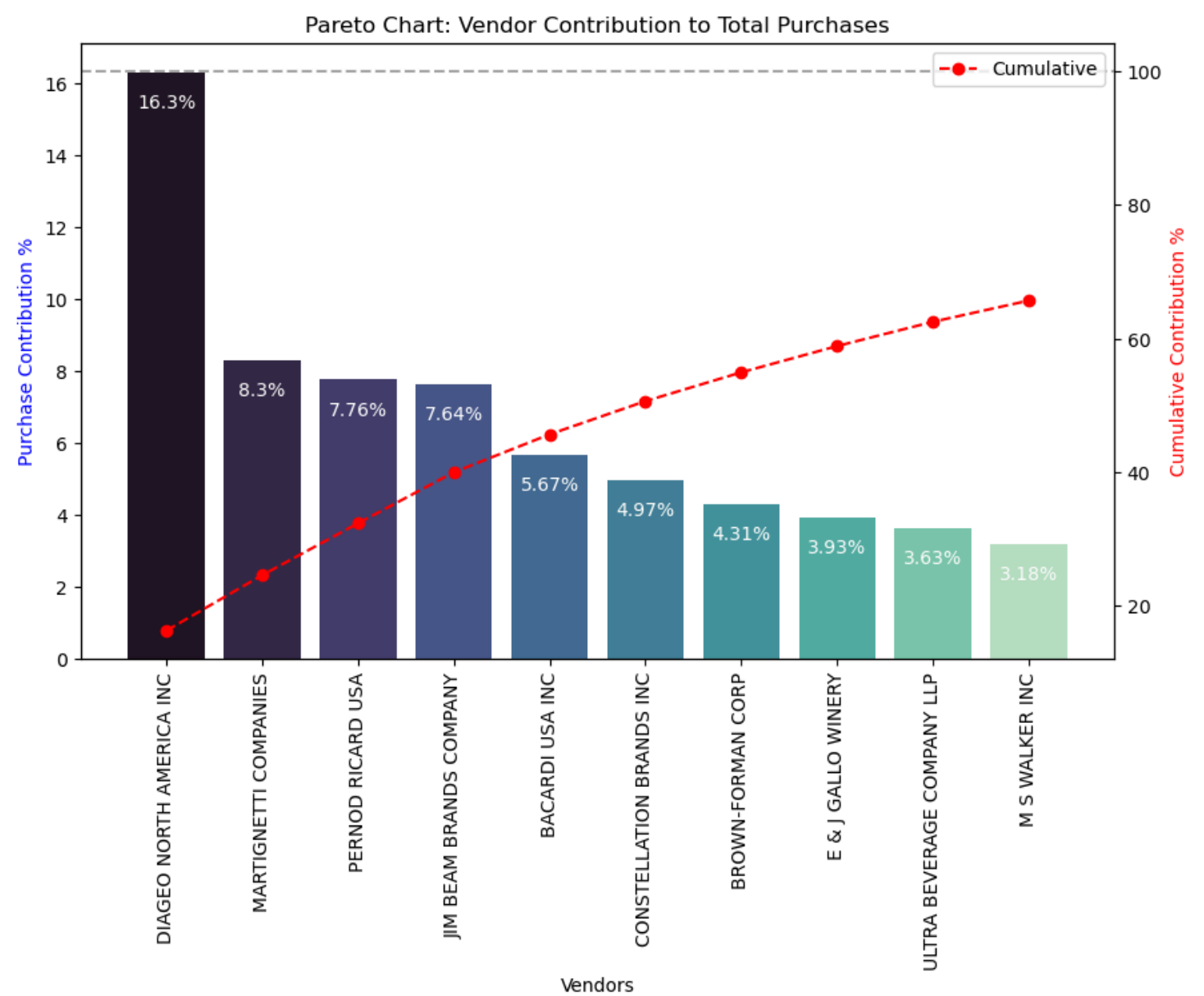
Top Brands by Total Sales

Rank	Brand	Sales
1	Jack Daniels No 7 Black	\$7.56M
2	Tito's Handmade Vodka	\$7.40M
3	Grey Goose Vodka	\$7.21M
4	Capt Morgan Spiced Rum	\$6.36M
5	Absolut 80 Proof	\$6.24M
6	Jameson Irish Whiskey	\$5.72M

Rank	Brand	Sales
7	Ketel One Vodka	\$5.07M
8	Baileys Irish Cream	\$4.15M
9	Kahlua	\$3.60M
10	Tanqueray	\$3.46M

What we saw: Top 10 vendors generate about 280Minsales, andthetop10brandsmakeabout57M.

6.3 Pareto view - vendor contribution



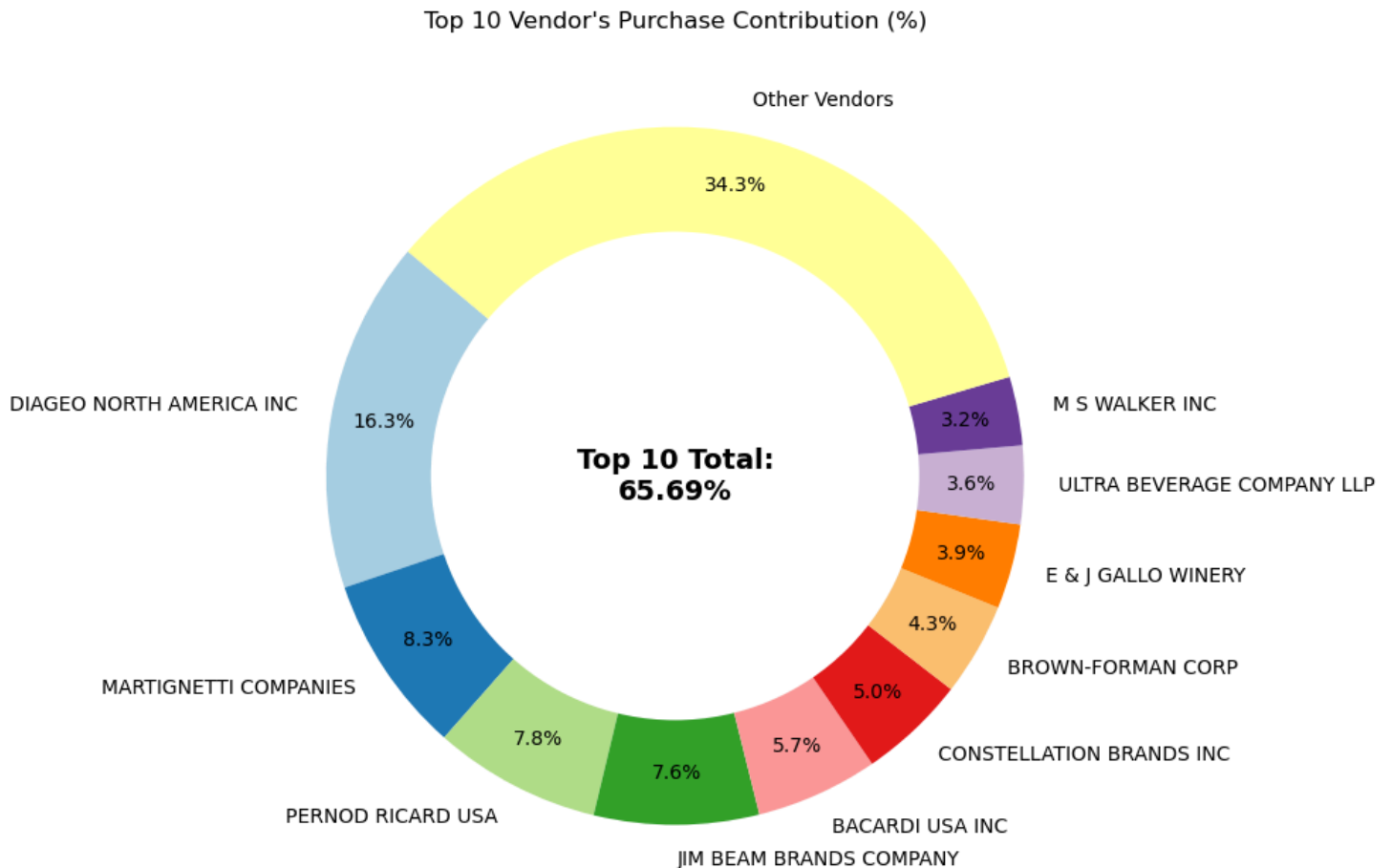
80/20 view:

- **Top Vendor** (DIAGEO NORTH AMERICA INC): 16.3% of all purchases

- **Top 2 Vendors:** 24.6% of purchases (16.3% + 8.3%)
- **Top 5 Vendors:** 45.3% of total purchases
- **Top 10 Vendors:** 65.7% of total purchases

This shows the vendor base is concentrated, so the top 5-10 vendors are especially important.

6.4 Market Share - Top 10 Vendors

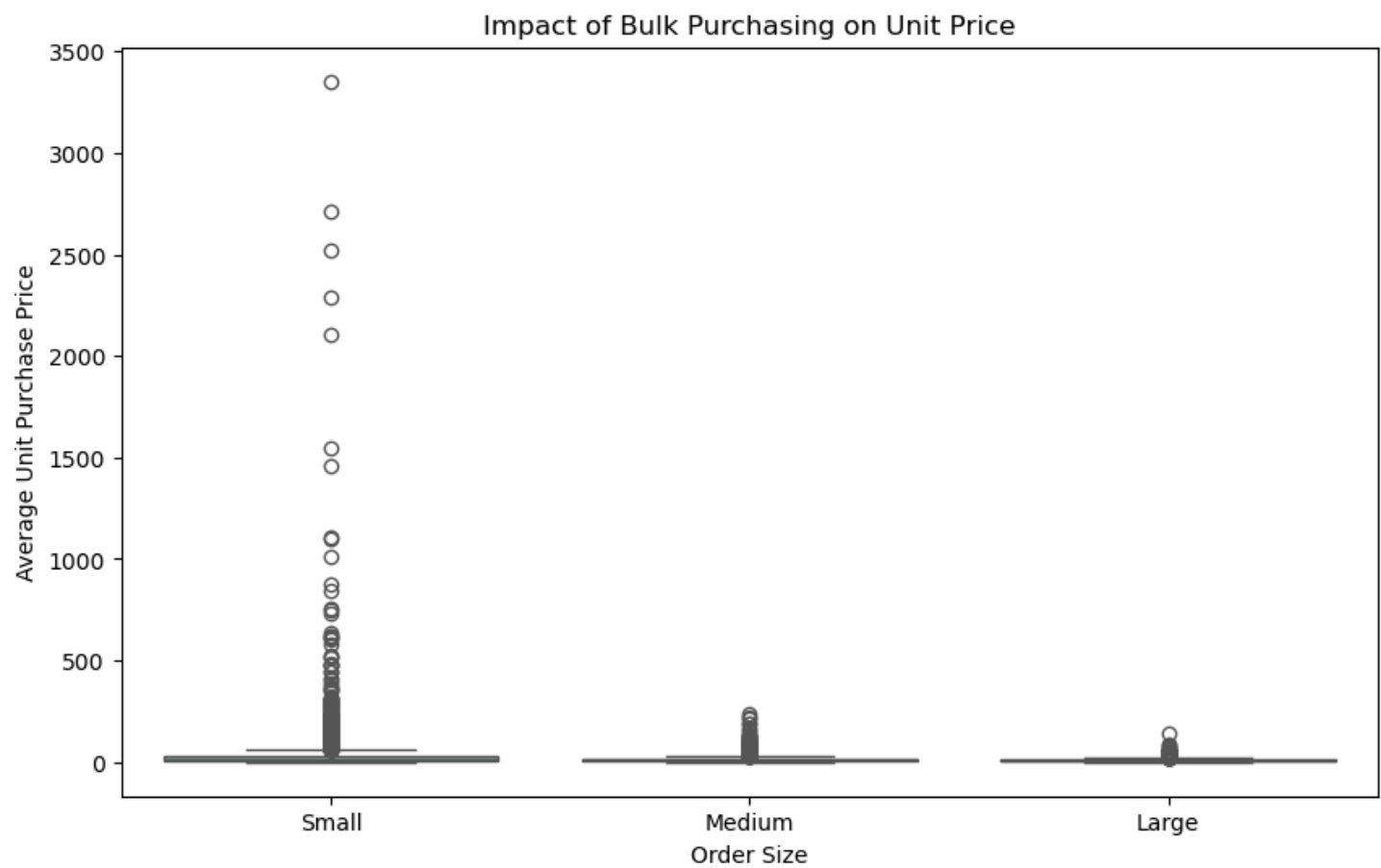


Detailed Market Share (Top 10):

- DIAGEO NORTH AMERICA INC: 16.3%
- MARTIGNETTI COMPANIES: 8.3%
- PERNOD RICARD USA: 7.8%
- JIM BEAM BRANDS COMPANY: 7.6%
- BACARDI USA INC: 5.7%
- CONSTELLATION BRANDS INC: 5.0%
- BROWN-FORMAN CORP: 4.3%
- E & J GALLO WINERY: 3.9%
- ULTRA BEVERAGE COMPANY LLP: 3.6%
- M S WALKER INC: 3.2%
- **Other Vendors:** 34.3% (rest of 100+)

Risk note: The top vendor supplies more than 16%, so it makes sense to keep other suppliers available.

6.5 Bulk Purchasing Impact on Unit Price



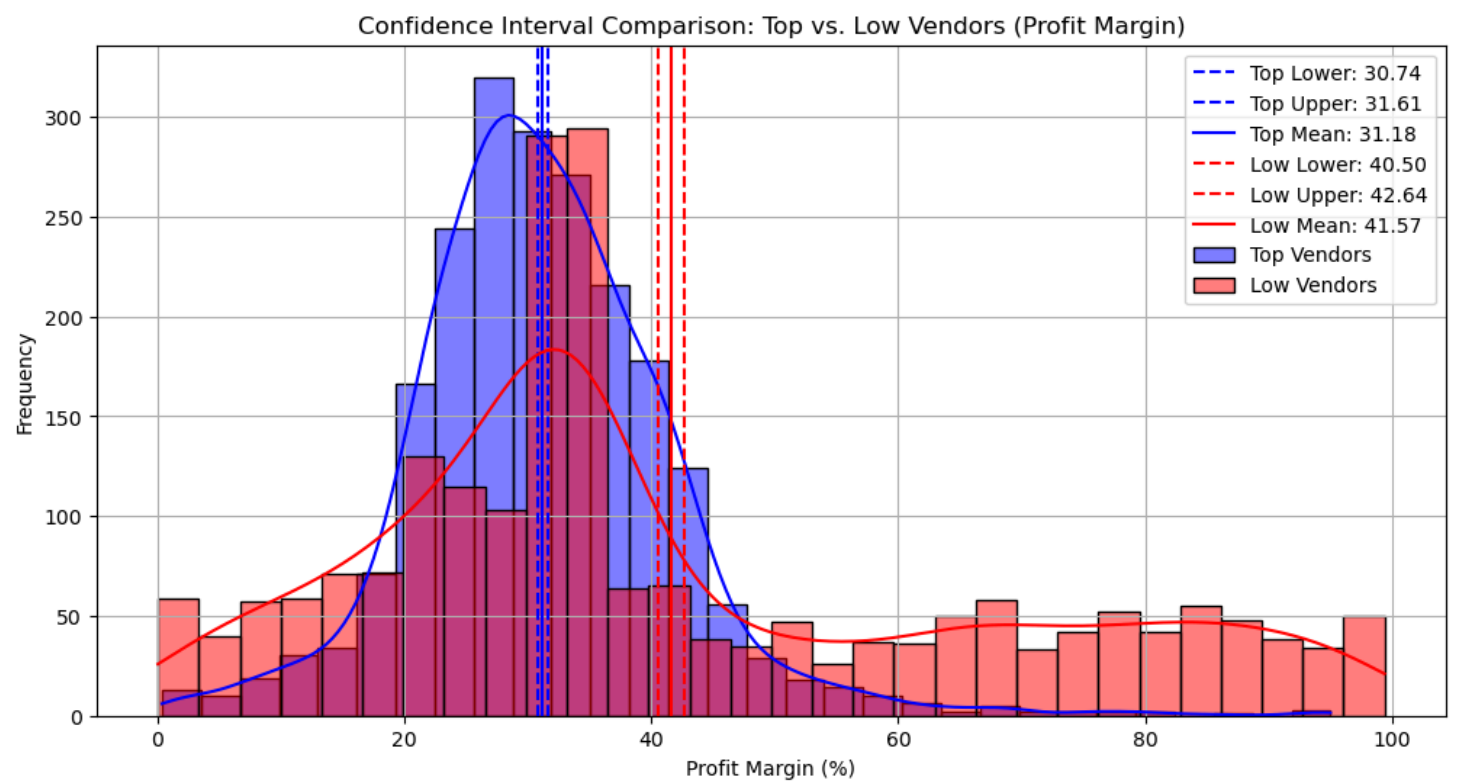
Order Size Categories:

- **Small Orders:** High purchase price variance (10—3,400 range)
 - Average unit purchase price: ~\$100+
- **Medium Orders:** Reduced variance, better price consistency
 - Average unit purchase price: ~\$50
- **Large Orders:** Lowest unit prices achieved
 - Average unit purchase price: ~10—50

Conclusion: Larger orders get much better unit prices than smaller ones.

Suggested action: Consider setting minimum order sizes or offering bulk purchase deals to cut procurement costs.

6.6 Comparison: Top vs. low vendors



Top vendors vs. low performers (profit margin):

Group	Mean Margin	95% CI Lower	95% CI Upper	Sample Size
Top Vendors	31.18%	30.74%	31.61%	~2,000
Low Performers	41.57%	40.50%	42.64%	~500

What this shows:

- Low-performing vendors actually have higher profit margins (41.57% vs. 31.18%)
- That likely happens because they sell less volume and avoid low-margin bulk sales
- The difference is clear and meaningful
- The biggest vendors may be accepting lower margins to sell more volume

What it means: Success is not just about margin; volume matters too.

7. Key Performance Indicators (KPIs)

Financial Metrics

KPI	Value	Benchmark	Status
Total Sales Revenue	\$452M+	-	Strong

KPI	Value	Benchmark	Status
Total Gross Profit	\$130M+	-	Strong
Average Profit Margin	30.4%	20-25% industry	Excellent
Avg Freight Cost	\$61,434	-	High (optimization opportunity)

Operational Metrics

KPI	Value	Assessment
Average Stock Turnover	0.98x	Near 1:1 (healthy balance)
Sales-to-Purchase Ratio	1.44x	Good; some fulfillment from existing stock
Product Variety	10,692 SKUs	High; enables broad market reach
Vendor Count	100+	Diversified; mitigates supply risk

8. Main findings and next steps

8.1 Vendor relationship notes

- ✓ **Finding:** Top 10 vendors represent 65.7% of all purchases
 - **What to do:** Set up partner tiers like platinum, gold, and silver based on volume
 - **Why it helps:** Gives better leverage on price and terms
- ✓ **Finding:** DIAGEO NORTH AMERICA INC dominates (16.3% market share)
 - **What to do:** Negotiate volume discounts and build a closer deal with the vendor
 - **Risk:** Relying too much on one supplier, so keep others ready

8.2 Profitability Optimization

- ✓ **Finding:** 75 high-margin products (>65% margin) with low sales (<\$500)
 - **What to do:** Run targeted promotions, like small price cuts or bundles
 - **Possible ROI:** 15-25% sales lift without cutting margin too much
- ✓ **Finding:** Profit margin positively correlates with sales volume (0.40 correlation)
 - **What to do:** Push high-margin products through volume-focused efforts

8.3 Operational Efficiency

✓ **Finding:** Economies of scale evident; large orders achieve 5-10x better unit pricing

- **What to do:** Combine small orders and use minimum order quantities
- **Possible savings:** 10-15% off the cost of goods sold

✓ **Finding:** Freight costs highly variable (0.09to257,032)

- **What to do:** Review logistics, consolidate shipments, and negotiate freight terms
- **Possible savings:** 5-10% on freight costs, or \$3-6M per year

8.4 Inventory Management

✓ **Finding:** Stock turnover averages 0.98x; some items exceed 100x (anomalies)

- **What to do:** Cut slow-moving SKUs, and check fast sellers for more stock or better pricing
- **Possible benefit:** Lower carrying costs and better cash flow

✓ **Finding:** 2,800+ products with zero or negative sales (losses)

- **What to do:** Consider clearance sales or talk to vendors about buybacks
- **Possible recovery:** \$10-20M in tied-up capital

8.5 Product focus

High Priority Brands (High volume + healthy margins):

- Jack Daniels No 7 Black
- Tito's Handmade Vodka
- Grey Goose Vodka
- Capt Morgan Spiced Rum
- Absolut 80 Proof
- **What to do:** Give these top brands more space, promotion, and vendor support

9. Recommendation summary

Priority	Initiative	Estimated benefit	Timeline
HIGH	Promote 75 high-margin, low-sales products	\$5-8M revenue increase	30-60 days
HIGH	Consolidate orders to improve unit pricing	\$10-15M cost reduction	60-90 days
HIGH	Audit and improve freight costs	\$3-6M savings	30-45 days

Priority	Initiative	Estimated benefit	Timeline
MEDIUM	Renegotiate top 10 vendor agreements	5-10% better terms	60-120 days
MEDIUM	SKU rationalization (discontinue slow movers)	\$10-20M capital recovery	90-180 days
MEDIUM	Set up a tiered vendor partnership program	Better collaboration	45-60 days
LOW	Premium product segment review	Useful insights	Ongoing

10. Conclusion

The vendor data shows a solid beverage business with:

- **Strengths:** strong margins (30%+), a broad vendor base, and good sales
- **Opportunities:** better promotions, lower costs, and less vendor concentration risk
- **Risks:** freight costs are uneven, inventory needs better control, and one vendor is too large a share

If the suggested actions are taken, the company could see:

- **More revenue:** \$5-8M
- **Lower costs:** \$13-21M
- **Total benefit:** \$18-29M (about a 5-8% margin lift)

Appendix: Visualizations Reference

Chart	Purpose	Location
Distribution Plots	Understand data spread and patterns	output_6_0.png
Box Plots	Identify outliers and anomalies	output_7_0.png
Cleaned Distributions	Post-filtering data quality	output_11_0.png
Categorical Counts	Vendor & product frequency	output_12_0.png
Correlation Matrix	Variable relationships	output_13_0.png
Promotional Opportunities	High-margin/low-sales view	output_22_0.png
Top 10 vendors and brands	Vendor and brand rankings	output_28_0.png
Pareto Chart	80/20 vendor concentration	output_36_0.png
Market Share Pie	Top 10 vendor contribution	output_39_0.png
Bulk Purchasing Impact	Order size vs. unit price	output_45_0.png
Confidence	Top vs. low vendor margins	output_57_1.png

Report Generated: May 2026

Data Period: Complete Inventory Cycle

Scope: 10,692 vendor-product transactions across 100+ vendors and 5,000+ brands